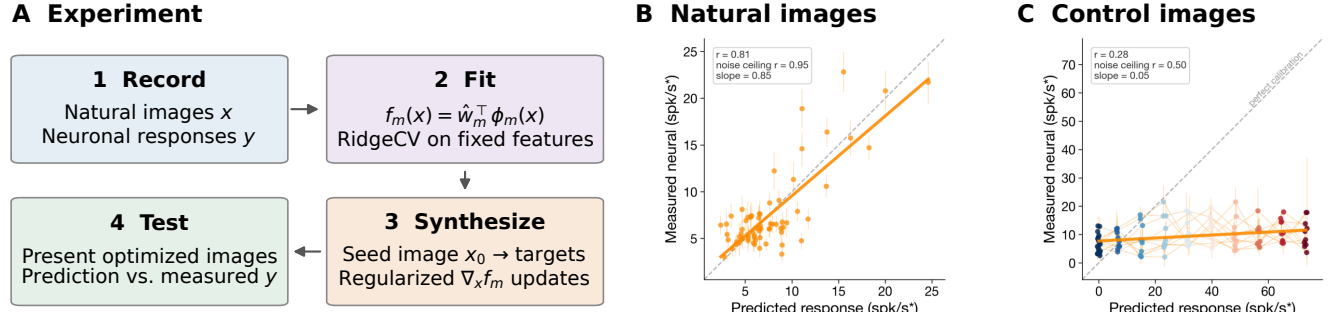

A Experiment
1 Record
Natural images x
Neuronal responses y
2 Fit
$f_m(x) = \hat{w}_m^\top \phi_m(x)$
RidgeCV on fixed features
3 Synthesize
Seed image x_0 → targets
Regularized $\nabla_x f_m$ updates
4 Test
Present optimized images
Prediction vs. measured y
B Natural images
r = 0.81
noise ceiling r = 0.95
slope = 0.85
Measured neural (spk/s*)
Predicted response (spk/s*)
C Control images
r = 0.28
noise ceiling r = 0.50
slope = 0.05
perfect calibration
Measured neural (spk/s*)
Predicted response (spk/s*)